Small Neural Networks for Handwritten Digits

MLP vs CNN vs Transformer on MNIST — and what image augmentation adds

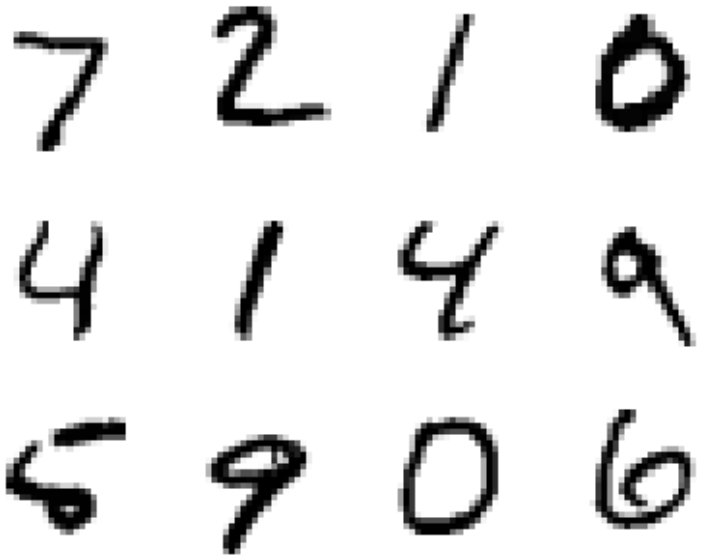
Best clean test accuracy: 99.59% (CNN)

AI homework · Fall 2026

Problem and data

Classify 28×28 grayscale digits into 10 classes

- MNIST: 60,000 training images, 10,000 test images.
- Strict split: train on the train part only; the test part is touched once, at the end.
- Last 5,000 training images held out as validation to pick the best epoch, so no decision ever looks at the test set.
- Same optimizer for all models: AdamW + one-cycle learning rate, label smoothing 0.05, batch 128.



sample test images

Three architectures

Same data, same training recipe

MLP

203,530 params

- Flatten 784 pixels
- 1 hidden layer (256, ReLU)
- Dropout → 10 classes
- No notion of neighbours

CNN

468,010 params

- 2 conv blocks (+BN, max-pool)
- 28×28 → 14×14 → 7×7
- Dense 128 → 10 classes
- Built-in locality + weight sharing

Transformer

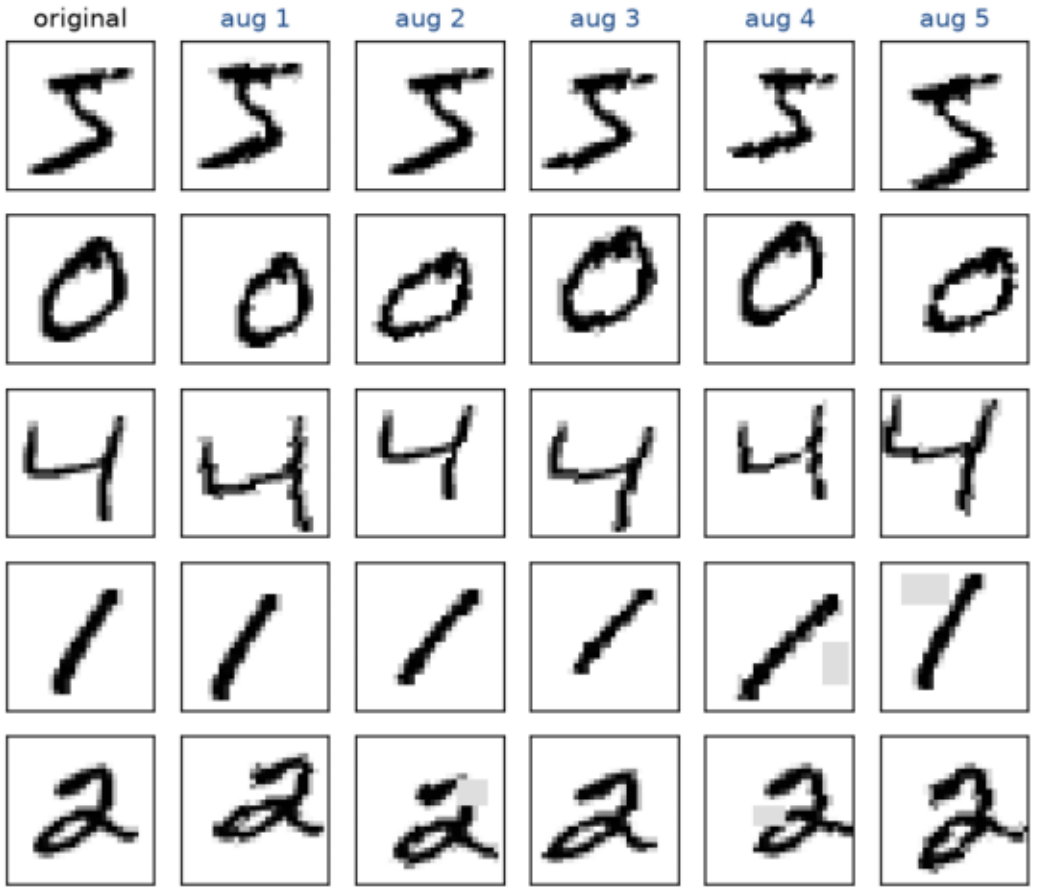
139,018 params

- 16 patches of 7×7 → 64-d tokens
- + [CLS] token, position emb.
- 4 encoder layers, 4 heads
- Attention between patches

Image augmentation

Random, label-preserving changes to each training image, every epoch

- Rotation up to ±12°, shift up to 12%, scale 0.88–1.12, shear ±8°.
- Random erasing of a small patch (25% of images).
- No flips — a mirrored 2 or 7 is no longer the same digit.
- Validation and test images are never augmented.
- Extra robustness test: the same test images rotated ±20°, shifted 15%, scaled 0.85–1.15.

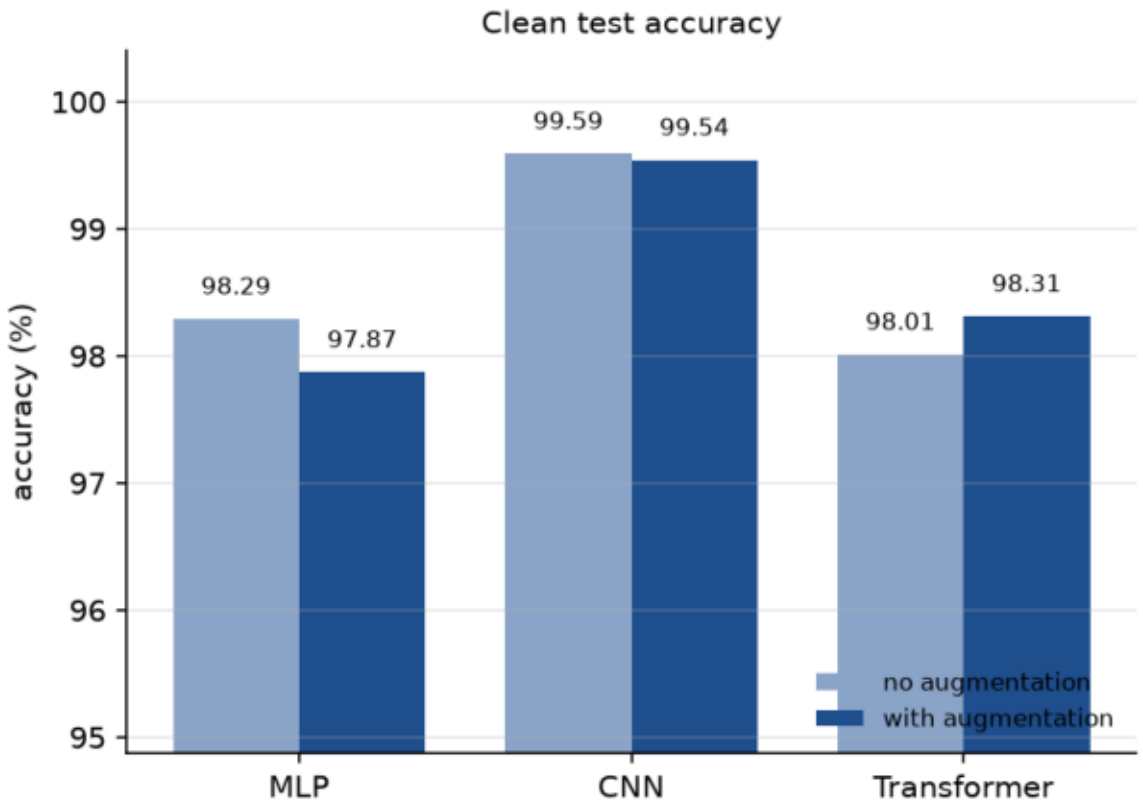


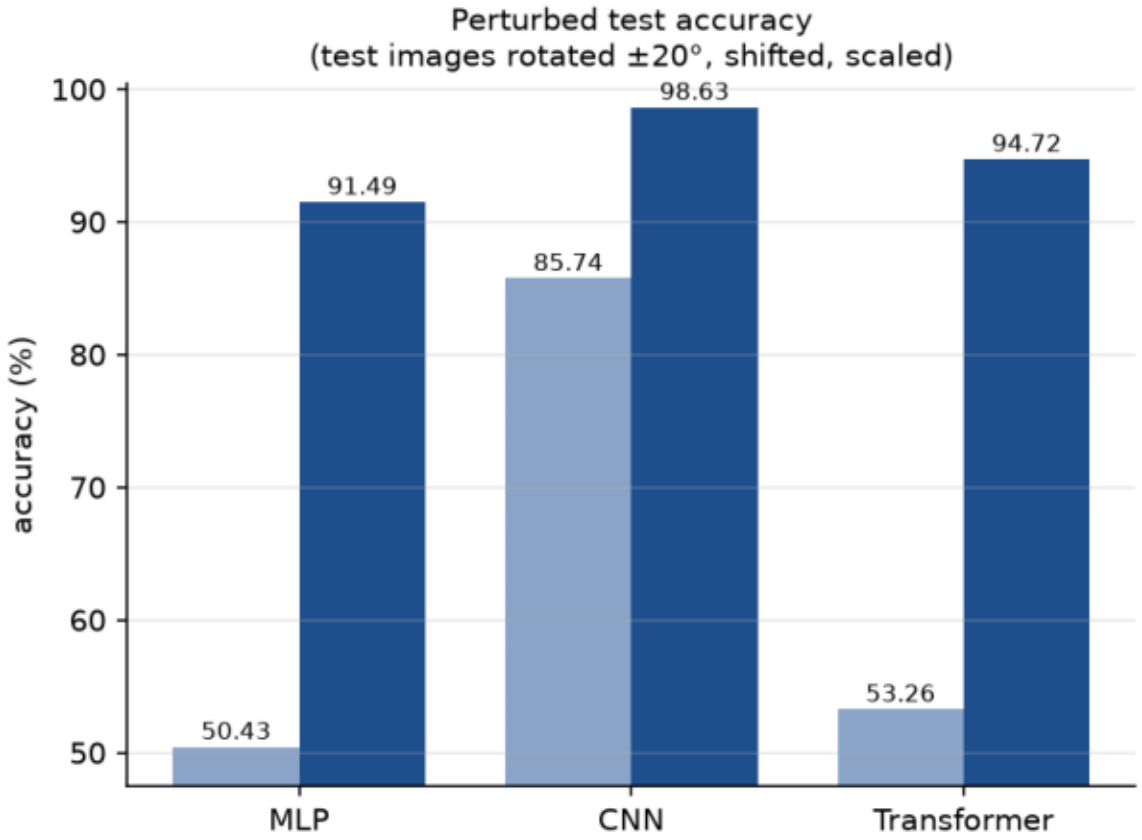
Results on the 10,000-image test set

Model	Augment	Params	Epochs	Train time	Test acc	Errors	Perturbed tes
MLP	no	203,530	10	54s	98.29%	171	50.43%
MLP	yes	203,530	10	86s	97.87%	213	91.49%
CNN	**no**	**468,010**	**8**	**527s**	**99.59%**	**41**	**85.74%**
CNN	yes	468,010	8	547s	99.54%	46	98.63%
Transformer	no	139,018	15	398s	98.01%	199	53.26%
Transformer	yes	139,018	15	439s	98.31%	169	94.72%

Device: cpu. Best epoch chosen by validation accuracy.

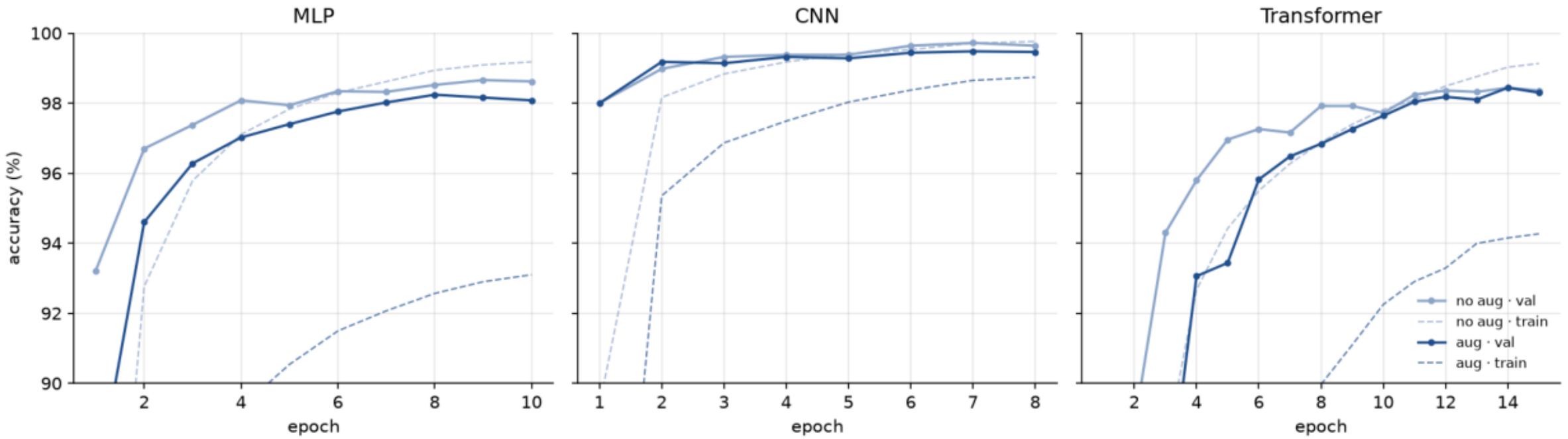
Clean vs perturbed accuracy





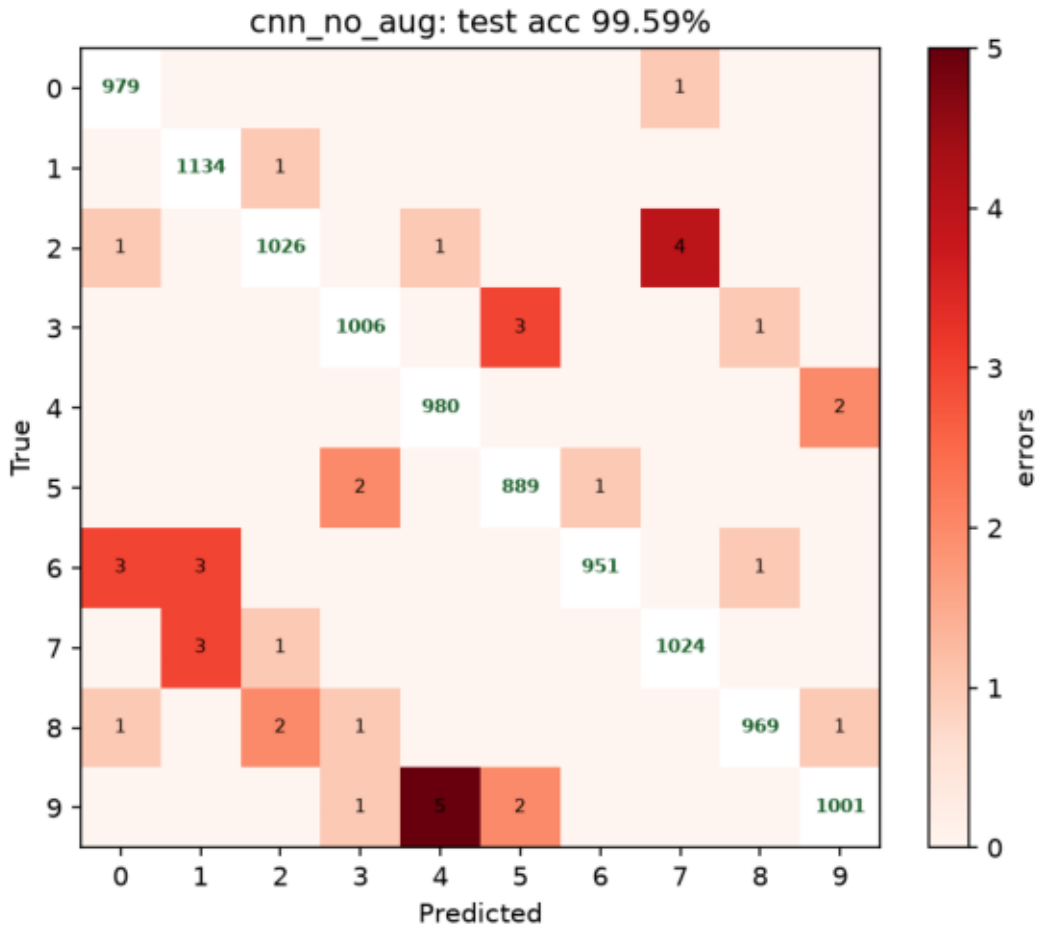
Training curves

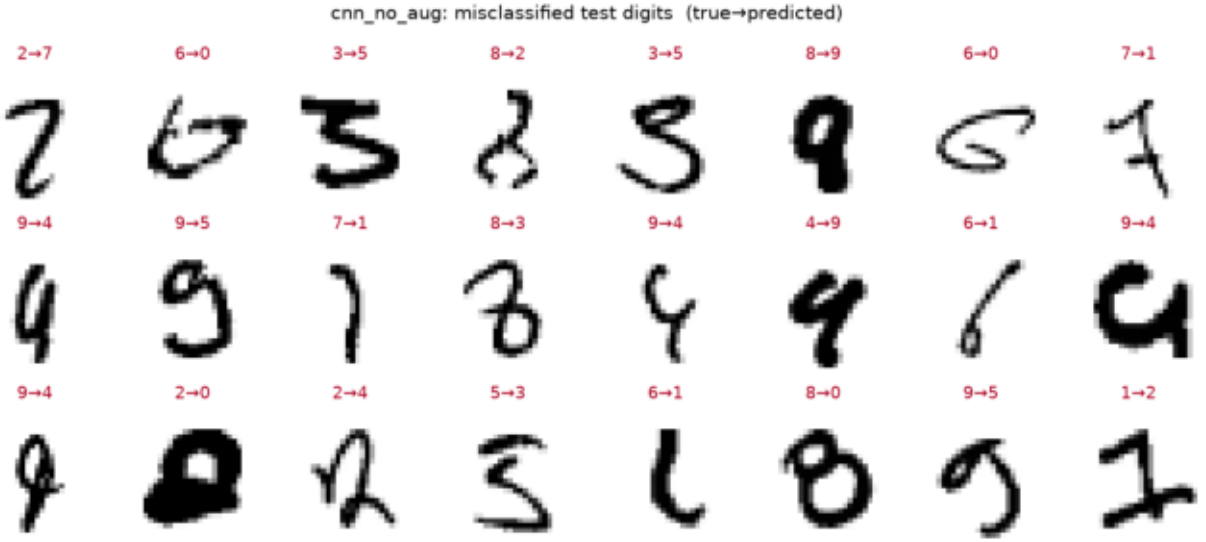
Solid = validation, dashed = training



Where the best model still fails

CNN: 41 mistakes out of 10,000





Takeaways

- Without augmentation: CNN 99.59% > MLP 98.29% > Transformer 98.01%.
- MLP: augmentation changed clean test accuracy by -0.42 pts and perturbed accuracy by +41.06 pts.
- CNN: augmentation changed clean test accuracy by -0.05 pts and perturbed accuracy by +12.89 pts.
- Transformer: augmentation changed clean test accuracy by +0.30 pts and perturbed accuracy by +41.46 pts.
- Best overall: CNN at 99.59% (41 errors).